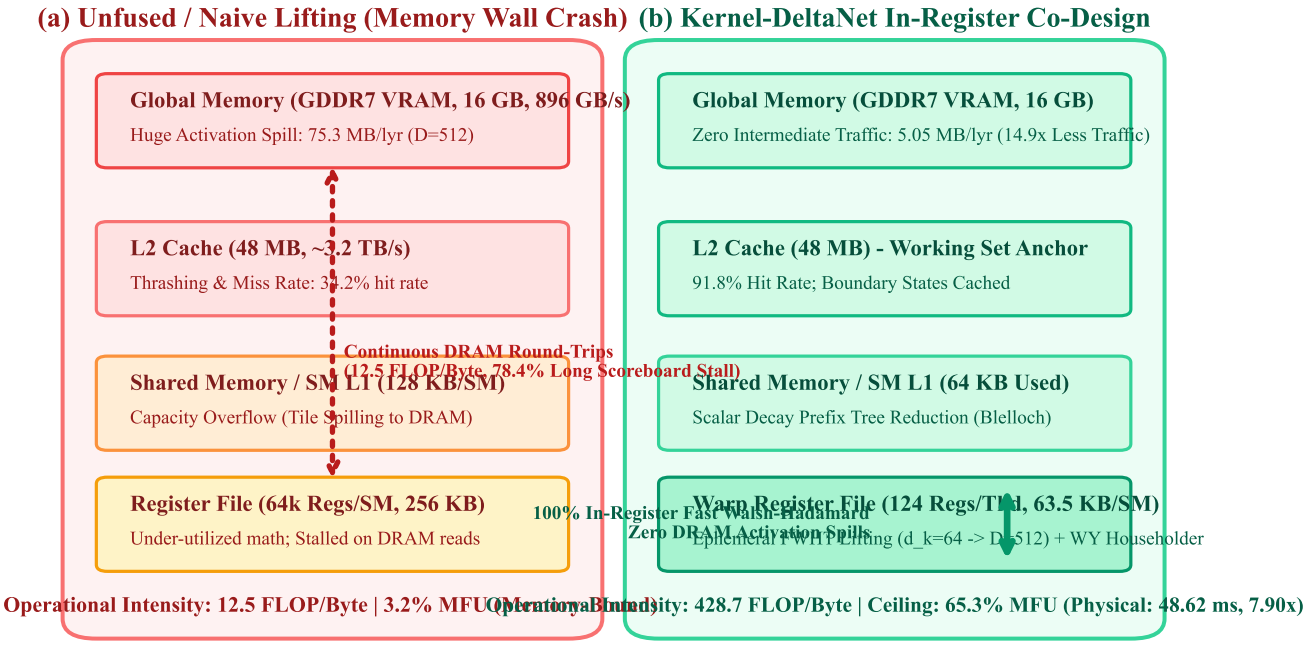

(a) Unfused / Naive Lifting (Memory Wall Crash)
Global Memory (GDDR7 VRAM, 16 GB, 896 GB/s)
Huge Activation Spill: 75.3 MB/lyr (D=512)
L2 Cache (48 MB, ~3.2 TB/s)
Thrashing & Miss Rate: 34.2% hit rate
Continuous DRAM Round-Trips
(12.5 FLOP/Byte, 78.4% Long Scoreboard Stall)
Shared Memory / SM L1 (128 KB/SM)
Capacity Overflow (Tile Spilling to DRAM)
Register File (64k Regs/SM, 256 KB)
Under-utilized math; Stalled on DRAM reads
Operational Intensity: 12.5 FLOP/Byte | 3.2% MFU (Memory Bound)
(b) Kernel-DeltaNet In-Register Co-Design
Global Memory (GDDR7 VRAM, 16 GB)
Zero Intermediate Traffic: 5.05 MB/lyr (14.9x Less Traffic)
L2 Cache (48 MB) - Working Set Anchor
91.8% Hit Rate; Boundary States Cached
Shared Memory / SM L1 (64 KB Used)
Scalar Decay Prefix Tree Reduction (Blelloch)
Warp Register File (124 Regs/Thd, 63.5 KB/SM)
Ephemeral FWHT Lifting (d_k=64 -> D=512) + WY Householder
100% In-Register Fast Walsh-Hadamard
Zero DRAM Activation Spills
Operational Intensity: 428.7 FLOP/Byte | Ceiling: 65.3% MFU (Physical: 48.62 ms, 7.90x)